

FORM TP 21225



TEST CODE **000574**

MAY/JUNE 2001

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

MATHEMATICS

UNIT 2 – PAPER 02

$2\frac{1}{2}$ hours

05 JUNE 2001 (a.m.)

This examination paper consists of **THREE** sections: Module 2.1, Module 2.2 and Module 2.3.

Each section consists of 2 questions.

The maximum mark for each section is 50.

The maximum mark for this examination is 150.

This examination consists of 5 pages.

INSTRUCTIONS TO CANDIDATES

1. **DO NOT** open this examination paper until instructed to do so.
2. Answer **ALL** questions from the **THREE** sections.
3. Unless otherwise stated in the question, all numerical answers **MUST** be given exactly **OR** to three significant figures as appropriate.

Examination Materials

Mathematical formulae and tables

Electronic calculator

Ruler and compass

SECTION A (MODULE 2.1)

Answer ALL questions.

1. (a) The parametric equations of a curve are given by $x = t^3 e^t$ and $y = t e^t$.

(i) Show that $\frac{dy}{dx} = \frac{t+1}{t^2(t+3)}$. [4 marks]

- (ii) Hence, state the values of t for which the derivative of the curve is not defined. [2 marks]

- (iii) Determine the equation of the tangent at the stationary point on the curve. [3 marks]

- (b) Using the derivatives of $\sin x$ and $\cos x$ with respect to x , show that the derivative of $\tan x$ with respect to x is $\sec^2 x$. [5 marks]

- (c) A curve has equation $y = \tan\left(x + \frac{\pi}{6}\right)$, where $0 < x < \frac{\pi}{3}$.

(i) Show that $\frac{dy}{dx} = 1 + y^2$. [3 marks]

- (ii) Hence, calculate the exact coordinates of the point on the curve at which the gradient is equal to 2. [6 marks]

- (iii) Find an expression for $\frac{d^2y}{dx^2}$ in terms of y . [2 marks]

Total 25 marks

2. (a) Find

(i) $\int 2x e^x dx$ [4 marks]

(ii) $\int \frac{x}{(x+4)(6-x)} dx$. [5 marks]

(b) The rate at which an air freshener block evaporates is directly proportional to its volume. At time, t weeks, the volume of the block is $V \text{ cm}^3$.

(i) Denoting the positive constant of proportionality by k , write down a differential equation relating V and t . [2 marks]

(ii) Hence, show that $V = Ae^{-kt}$, where A is a constant. [4 marks]

(iii) Given that A and k are both positive, sketch a graph showing the variation of V with t . [2 marks]

(iv) Initially, the air freshener block has volume 64 cm^3 . It loses half its volume after 6 weeks. Calculate the exact values of k and A . [5 marks]

(v) The air freshener block becomes ineffective when its volume reaches 6 cm^3 . Calculate the time, to the nearest week, at which the block should be replaced. [3 marks]

Total 25 marks

SECTION B (MODULE 2.2)

Answer ALL questions.

3. (a) (i) Prove, by mathematical induction, that

$$\sum_{r=1}^n r^3 = \frac{n^2(n+1)^2}{4} \quad [8 \text{ marks}]$$

- (ii) Deduce that $\sum_{r=1}^n (2r)^3 = 2n^2(n+1)^2$. [2 marks]

- (b) Given that the sum of the first n terms of a series, S , is $9 - 3^{2-n}$

- (i) find the n^{th} term of S [3 marks]

- (ii) show that S is a geometric progression [3 marks]

- (iii) find the sum to infinity of S . [2 marks]

- (c) (i) Use the fact that $\frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)}$ to show that

$$\sum_{r=1}^n \left(\frac{1}{k(k+1)} \right) = 1 - \frac{1}{n+1} \quad [4 \text{ marks}]$$

- (ii) Deduce that $\sum_{r=1}^{\infty} \left(\frac{1}{k(k+1)} \right) = 1$. [3 marks]

Total 25 marks

4. (a) Explain carefully and in your own words, the Newton-Raphson method for numerically determining the root of the equation $f(x) = 0$, where $f(x)$ is a continuous, differentiable function. [9 marks]

- (b) (i) Sketch the functions, $y = \sin x$ and $y = x^2$, on the same axes and state why the function, $f(x) = \sin x - x^2$, has exactly two real roots. [4 marks]

- (ii) Construct a table for the function, $f(x) = \sin x - x^2$, for $x = 0, 0.25, 0.5, 0.75, 1$. [3 marks]

- (iii) Give the interval in which the non-zero root, α , of $f(x)$ lies. [3 marks]

- (iv) Starting with a first approximation of α , find a better approximation. [3 marks]

- (v) Calculate the value of α to 3 decimal places. [3 marks]

Total 25 marks

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SECTION C (MODULE 2.3)

Answer ALL questions.

5. (a) A die, numbered 1 to 6 is biased in such a way that a 6 is twice as likely as any other score. Find $P(6)$, the probability that a six shows when this die is thrown. [3 marks]
- (b) Three dice, two of which are biased as in 5 (a) above and the other fair, are thrown.
- (i) Draw a tree diagram to represent the throwing of the three dice showing the probabilities of sixes and not sixes. [12 marks]
- (ii) Find the probability that exactly one 6 turns up. [5 marks]
- (iii) Find the probability that when one 6 turns up it appears on the fair die. [5 marks]

Total 25 marks

6. At the start of a month, a customer owes a credit card company \$1000.00. In the middle of the month, the customer pays \$P to the company, where $P < 1000.00$, and at the end of the month, the company adds interest at the rate, R, of 3% of the amount still owing. This process is repeated in each subsequent month.
- (a) Find the value of P for which the customer owes \$1000.00 at the start of every month. [6 marks]
- (b) Find the value of P for which the whole amount is paid off after the second payment. [6 marks]
- (c) Assuming that the debt has not been paid off after 4 payments, show that the amount still owing at the beginning of the 5th month can be expressed as

$$\left[1000R^4 - \frac{PR(R^4 - 1)}{R - 1} \right], R = 1.03. \quad [6 \text{ marks}]$$

- (d) Show that the value of P for which the whole amount owing is exactly paid off after the nth payment is given by

$$P = \frac{1000 R^n - 1(R - 1)}{R^n - 1}. \quad [7 \text{ marks}]$$

Total 25 marks

END OF TEST